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DUAL-TRAP TOILET HAVING IMPROVED TRAPWAY AND IMPROVED CONNECTING TUBE INTERFACE

Abstract

A toilet assembly may comprise a toilet tank to hold flush water, a flush valve assembly positioned in the toilet tank, a toilet deck, a toilet bowl, a trapway in flow communication with the toilet bowl, a container positioned in the toilet tank, and a connecting tube extending from an interior of the container to the trapway. A first gasket positioned between the toilet tank and the toilet deck may form a seal around the flush valve assembly. A second gasket positioned between the toilet tank and the toilet deck may form a seal around the connecting tube. The first gasket may comprise a recess for receiving the flush valve assembly and a protrusion extending into the toilet deck.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/333,971, filed Apr. 22, 2022, U.S. Provisional Application No. 63/333,973, filed Apr. 22, 2022, and U.S. Provisional Application No. 63/333,977, filed Apr. 22, 2022, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates generally to dual-trap toilets.

BACKGROUND

[0003] Conventional dual-trap toilets comprise a trapway that is configured to trap water in two separate locations. A first upstream trap may be located directly downstream of the toilet bowl and may be configured to hold water in the toilet bowl in between flush cycles. A second downstream trap may be located between the first trap and an outlet from the trapway to a drainage pipe. The dual-trap configuration may prevent sewer gases from leaking out of the sewer through drainage pipe while also providing a quieter flush cycle than other flush toilets (e.g., single trap toilets).

SUMMARY

[0004] The tank of a dual-trap toilet may comprise one or more components that are fluidly coupled to the toilet bowl and/or to the trapway. The interface between the one or more components and a toilet deck above the toilet bowl and/or trapway may be configured to prevent flush water and air from leaking out of the toilet tank and into other portions of the dual-trap toilet. The present disclosure provides dual-trap toilet assemblies having an improved interface between the trapway, the toilet bowl, and the toilet tank.

[0005] A dual-trap toilet may have one or more gaskets that are configured to provide an effective seal between components of the toilet. Each of the one or more gaskets may comprise a collared portion and/or a recessed portion that protrudes into the toilet deck of the dual-trap toilet. Use of a gasket having a recess and/or protrusion as described herein may allow for the gasket to be compressed by forces acting in multiple different directions—e.g., the collar may be compressed vertically between the tank and the deck, while the protrusion may be compressed horizontally by a receiving area into which it is inserted. Compression by forces acting in multiple different directions may allow for a tighter and more secure fit to be achieved, and for a more effective and robust seal to be formed.

[0006] A first toilet assembly may comprise a toilet tank to hold flush water, a flush valve assembly positioned in the toilet tank, a toilet deck, a toilet bowl, a trapway in flow communication with the toilet bowl, a container positioned in the toilet tank, a connecting tube extending from an interior of the container to the trapway, a first gasket positioned between the toilet tank and the toilet deck and forming a seal around the flush valve assembly, and a second gasket positioned between the toilet tank and the toilet deck and forming a seal around the connecting tube. The container may be in flow communication with the toilet tank and may have an open lower end and a closed upper end. The trapway may comprise a sump trap, a first upstream weir, a lower trap, and a second downstream weir. The connecting tube may be coupled to the trapway at a position between the

sump trap and the lower trap and may provide flow communication between the container and the trapway.

[0007] In some embodiments of the first toilet assembly, the first gasket comprises a recess receiving the flush valve assembly.

[0008] In some embodiments of the first toilet assembly, the first gasket comprises a protrusion extending into the toilet deck.

[0009] In some embodiments of the first toilet assembly, the first gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion.

[0010] In some embodiments of the first toilet assembly, the second gasket comprises a recess receiving the connecting tube.

[0011] In some embodiments of the first toilet assembly, the second gasket comprises a protrusion extending into the toilet deck.

[0012] In some embodiments of the first toilet assembly, the second gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion.

[0013] In some embodiments of the first toilet assembly, between flush cycles, the container comprises pressurized air positioned above flush water.

[0014] In some embodiments of the first toilet assembly, upon discharging water into a flush valve core to initiate a flush cycle, reduced pressure is created in pressurized air in the trapway.

[0015] In some embodiments of the first toilet assembly, upon re-filling the toilet tank with water to end a flush cycle, increased pressure is created in pressurized air in the trapway.

[0016] In some embodiments of the first toilet assembly, the flush valve assembly comprises a valve body extending from a valve inlet to a valve outlet and a valve cover having a seal to enclose the valve inlet.

[0017] In some embodiments of the first toilet assembly, the container comprises a continuous side wall, an upper end wall, and an inner wall that together bound a vacuum chamber of the container.

[0018] In some embodiments of the first toilet assembly, the inner wall bounds an inner chamber of the container.

[0019] In some embodiments of the first toilet assembly, the second gasket comprises a face seal.

[0020] In some embodiments of the first toilet assembly, the second gasket comprises a piston seal.

[0021] A second toilet assembly may comprise a toilet tank to hold flush water, a flush valve assembly positioned in the toilet tank, a toilet deck, a toilet bowl, a trapway in flow communication with the toilet bowl, a container positioned in the toilet tank, a connecting tube extending from an interior of the container to the trapway, and a first gasket positioned between the toilet tank and the toilet deck and forming a seal around the flush valve assembly, wherein the first gasket comprises a recess receiving the flush valve assembly and a protrusion extending into the toilet deck. The container may be in flow communication with the toilet tank and may have an open lower end and a closed upper end. The trapway may comprise a sump trap, a first upstream weir, a lower trap, and a second downstream weir. The connecting tube may be coupled to the trapway at a position between the sump trap and the lower trap and may provide flow communication between the container and the trapway.

[0022] In some embodiments of the second toilet assembly, the first gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion.

[0023] In some embodiments of the second toilet assembly, an outer diameter of the protrusion of the first gasket is wider than an inner diameter of the recess of the first gasket.

[0024] In some embodiments, the second toilet assembly comprises a second gasket positioned between the toilet tank and the toilet deck and forming a seal around the connecting tube. The second gasket may comprise a recess receiving the connecting tube and a protrusion extending into the toilet deck.

[0025] In some embodiments of the second toilet assembly, the second gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion of the second gasket.

[0026] In some embodiments of the second toilet assembly, an outer diameter of the protrusion of the second gasket is wider than an inner diameter of the recess of the second gasket.

[0027] In some embodiments of the second toilet assembly, the second gasket comprises a face seal.

[0028] In some embodiments of the second toilet assembly, the second gasket comprises a piston seal.

[0029] In some embodiments of the second toilet assembly, between flush cycles, the container comprises pressurized air positioned above flush water.

[0030] In some embodiments of the second toilet assembly, upon discharging water into a flush valve core to initiate a flush cycle, reduced pressure is created in pressurized air in the trapway.

[0031] In some embodiments of the second toilet assembly, upon re-filling the toilet tank with water to end a flush cycle, increased pressure is created in pressurized air in the trapway.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0032] The following figures show various views and/or associated data for dual-trap toilets, in accordance with some embodiments. The dual-trap toilets shown in the figures may, in some embodiments, have any one or more of the characteristics described herein.

[0033] FIGS. 1A and 1B show perspective views of a dual-trap toilet, in accordance with some embodiments.

[0034] FIG. 2 shows a cross-sectional view of a dual-trap toilet, in accordance with some embodiments.

[0035] FIG. 3 shows a perspective cutaway view of a dual-trap toilet, in accordance with some embodiments. FIG. 4 shows a partial cross-sectional view of a dual-trap toilet, in accordance with some embodiments.

[0036] FIGS. 5A-5E show various views of components of a dual-trap toilet for connecting a connecting tube between a toilet tank and a trapway, in accordance with some embodiments.

[0037] FIGS. 6A-6E shows cross-sectional views of an interface between a toilet tank and a toilet deck, showing a valve gasket and a vacuum gasket, in accordance with some embodiments.

[0038] FIGS. 7A-7B show top and bottom views of a manifold plate, according to some embodiments.

DETAILED DESCRIPTION

[0039] Dual-trap toilets comprise a trapway that is configured to trap water in two separate locations. In some embodiments, dual-trap toilets may have a trapway including a sump trap, a first upstream weir, a lower trap, and a second downstream weir. A container positioned in the toilet tank may be coupled by a connecting tube that extends between an interior of the container and the trapway. The connecting tube may be coupled to the trapway at a position between the sump trap and the lower trap and may provide flow communication between the container and the trapway. Between flush cycles, the container and the trapway may contain positively pressurized air. The positively pressurized air in the trapway between flush cycles may push upwards on water in the sump trap and may result in a larger water spot in the toilet bowl than would be present if not for the positively pressurized air. Additionally, the positively pressurized air in the trapway between flush cycles may push downwards on water in the lower trap, such that a water level in an upstream end of the lower trap may be lower than a water level in a downstream end of the lower trap.

[0040] Upon initiation of a flush cycle, a flush valve seal may lift off of a flush valve inlet to the flush valve to open the flush valve, and flush water may be discharged from the toilet tank and the container through the flush valve and into the toilet bowl. As used herein, the term “flush water” may refer to any water that passes out of a toilet tank (or any subcomponent thereof, such as a

container positioned inside a toilet tank, an inner chamber of said container, and/or a vacuum chamber of said container) during a flush cycle. Discharging the flush water during a flush cycle may exert a negative pressure on air in an upper end of the container, connecting tube, and trapway portion between the sump trap and the lower trap. The negative pressure may cause a drop to atmospheric pressure or to partial vacuum. The negative pressure may help create a siphon to pull water and waste through the sump area and into and out of the trapway.

[0041] A flush cycle may be considered completed upon closing the flush valve and re-filling the toilet tank, sump trap, and lower trap. Upon completion of a flush cycle, new flush water entering the toilet tank may also enter the container via one or more openings positioned in the container wall. Entry of water into the container may compress air into an upper end of the container and may return the air contained in the regions defined by the upper end of the container, the connecting tube, and the trapway portion between a sump trap and a lower trap to atmospheric pressure and/or to a positive pressure above atmospheric pressure.

[0042] A dual-trap toilet may have one or more gaskets that are configured to provide an effective seal between components of the toilet. One or more of the gaskets may be positioned between a toilet tank and a toilet deck of the toilet. The gaskets may comprise one or more flexible and/or compressible materials, such as rubber or any natural or synthetic elastomer.

[0043] A first gasket may provide a seal around a flush valve assembly. The first gasket may form a seal around a portion of the flush valve assembly through which flush water from the flush valve may pass from the toilet tank into the toilet bowl. The seal may prevent flush water from leaking. A second gasket may provide a seal around a connecting tube. The second gasket may form a seal around a portion of the conduit through which pressurized air may pass between the container and the trapway. The seal may prevent pressurized air and/or vacuum from leaking. The seal that is formed between connecting tube, tank, and toilet deck may allow for a vacuum to be held in container upper end, connecting tube, and trapway during flush cycles; and/or may allow for pressurized air to be held in the container upper end, connecting tube, and trapway between flush cycles. Providing the two gaskets separately from one another, rather than providing a single gasket with multiple through-holes intended to provide both seals, may allow for the preciseness of the fit and the effectiveness of the seal of each of the gaskets to be improved.

Dual-Trap Toilets

[0044] Dual-trap toilets comprise a trapway that is configured to trap water in two separate locations. A first upstream trap may be located directly downstream of the toilet bowl and may be configured to hold water in the toilet bowl in between flush cycles. A second, downstream trap may be located between the first trap and an outlet from the trapway to a drainage pipe. The dual-trap configuration may prevent sewer gases from leaking out of the sewer through drainage pipe while also providing a quieter flush cycle than other flush toilets (e.g., single trap toilets).

[0045] FIGS. 1A and 1B show perspective views of a dual-trap toilet, in accordance with some embodiments. Specifically, FIGS. 1A and 1B illustrate perspective views of a dual-trap toilet **100** comprising a toilet bowl **102**, a toilet tank **104**, and a trapway **106**. In some embodiments, toilet bowl **102**, toilet tank **104**, and trapway **106** may be in flow communication. In some embodiments, as shown in FIG. 1B, toilet **100** may comprise a flush control **108**. Flush control **108** may comprise one or more handles, push buttons, levers, chains, position sensors, and/or motion sensors that are configured to initiate a flush cycle when engaged by a user.

[0046] Between flush cycles, toilet bowl **102** may contain a volume of water. In some embodiments, a volume of water contained in toilet bowl **102** between flush cycles may be about 0.5 gallons, about 0.75 gallons, about 1 gallon, about 1.25 gallons, about 1.5 gallons, about 1.75 gallons, about 2 gallons, about 3 gallons, about 4 gallons, or about 5 gallons. In some embodiments, a volume of water contained in toilet bowl **102** between flush cycles may be greater than or equal to 2.5 gallons, 3.5 gallons, 4.5 gallons, 5.5 gallons, 6.5 gallons, 7.5 gallons, or 8.5 gallons. In some embodiments, a volume of water contained in toilet bowl **102** between flush

cycles may be less than or equal to 2 gallons, 1.5 gallons, 1 gallon, 0.5 gallons, or 0.1 gallons. [0047] Toilet tank **104** may be configured to contain flush water. Toilet tank **104** may be configured to fluidly connect to a water supply. In some embodiments, when a flush cycle is initiated, toilet tank **104** may be configured to discharge a volume of contained flush water into toilet bowl **102**. Toward the end of a flush cycle, toilet tank **104** may be configured to replenish the discharged volume of water by receiving water from the water supply.

[0048] When a flush cycle is initiated (e.g., by a user who engages flush control **108**), the contents of toilet bowl **102** may be siphoned into trapway **106**. Trapway **106** may be configured to fluidly connect to a sewer system (or to one or more drain pipes that connect to a sewer system). During a flush cycle, the siphoned contents of toilet bowl **102** may be directed through trapway **106** and into the sewer system.

[0049] Trapway **106** may comprise a sump trap and a lower trap. The sump trap may be directly downstream of toilet bowl **102** and the lower trap may be downstream of the sump trap. When toilet **100** is between flush cycles, water may collect in both the sump trap and the lower trap. The collection of water in the sump trap and the lower trap may form a series of water seals which prevent leakage of sewer gas out of trapway **106**. The water that collects in the sump trap between flush cycles may be the volume of water contained in toilet bowl **102** between flush cycles, as described above.

[0050] When toilet **100** is between flush cycles, toilet tank **104** and trapway **106** may be configured to contain positively pressurized air. The positively pressurized air contained in trapway **106** may be contained between a sump trap and a lower trap and may exert upward pressure on water collected in the sump trap and downward pressure on water collected in the lower trap. In some embodiments, the upward pressure exerted on the water collected in the sump trap by the pressurized air contained in trapway **106** may raise the water level of water contained in toilet bowl **102** between flush cycles. The raised water level of water contained in toilet bowl **102** may reduce or prevent soiling of toilet bowl **102**.

[0051] In some embodiments, when a flush cycle is initiated and flush water is discharged from toilet tank **104** into toilet bowl **102**, a negative pressure may be exerted on the air contained in toilet tank **104** and the air contained in trapway **106**. The negative pressure may cause the pressure within toilet tank **104** and/or trapway **106** to decrease to atmospheric pressure and/or to a partial vacuum. This decrease in pressure may generate a siphon in trapway **106** that may cause the contents of toilet bowl **102** to be pulled into and subsequently out of trapway **106**.

[0052] Toward the end of a flush cycle, water may again collect in toilet bowl **102** and trapway **106**. Upon completion of a flush cycle, toilet tank **104** may be configured to receive new flush water from a water supply. As toilet tank **104** refills with flush water, the air contained in toilet tank **104** may be compressed, thereby increasing the pressure of the air contained in toilet tank **104** and/or the air contained in trapway **106** to atmospheric pressure and/or to a pressure above atmospheric pressure. When the air contained in toilet tank **104** and/or the air contained in trapway **106** return(s) to positively pressurized states, the flush cycle may be completed.

[0053] FIG. 2 shows a cross-sectional view of a dual-trap toilet, in accordance with some embodiments. Specifically, FIG. 2 illustrates a cross-sectional view of a toilet **200** comprising a toilet bowl **202**, a toilet tank **204**, a trapway **206**, and a flush control **208**. In some embodiments, toilet bowl **202**, toilet tank **204**, trapway **206**, and/or flush control **208** may include features of toilet bowl **102**, toilet tank **104**, trapway **106**, and/or flush control **108** shown in FIG. 1A and/or FIG. 1B.

[0054] A container **218** and a flush valve **224** may be positioned within and in flow communication with toilet tank **204**. Container **218** may have an irregular toroidal structure that defines a vacuum chamber, an inner chamber, and an outer region in toilet tank **204** that is external to and surrounds both container **218** and the inner chamber. In some embodiments, flush valve **224** may be positioned in the inner chamber defined by container **218** such that container **218** encircles flush valve **224**.

[0055] Between flush cycles, toilet tank **204** may be configured to contain flush water. Between flush cycles, a first volume of flush water may be contained the outer region in toilet tank **204** that is external to and surrounds container **218** and the inner chamber defined by the structure of container **218**, a second volume of flush water may be contained in the central cavity region defined by the structure of container **218**, and a third volume of flush water may be contained in a vacuum chamber contained within the internal irregular annular region of container **218**.

[0056] Toilet bowl **202** may be fluidly connected to toilet tank **204** via a fluid pathway **222**. Flush valve **224** may be in flow communication with fluid pathway **222** which may, in turn, be in flow communication with toilet bowl **202**. In some embodiments, a flush valve gasket **210** may be positioned at an interface between flush valve **224** and fluid pathway **222**. Flush valve gasket **210** may form a mechanical seal between flush valve **224** and fluid pathway **222** in order to prevent flush water and/or air from leaking out of toilet tank **204** and into toilet bowl **202** between flush cycles (i.e., to prevent water and/or air leakage from toilet tank **204** when flush valve **224** is closed).

[0057] Fluid pathway **222** may be in flow communication with a rim channel **236** that encircles the rim of toilet bowl **202**. During a flush cycle, a portion of flush water may be transmitted from toilet tank **204**, through fluid pathway **222**, and into rim channel **236**. The flush water may then be transmitted into toilet bowl **202** from rim channel **236** through one or more rim outlets.

[0058] The portion of flush water transmitted from toilet tank **204**, through fluid pathway **222**, and into rim channel **236** may be between about 0.5-2 liters, about 1-3 liters, about 1-4 liters, or about 1-5 liters. Optionally, the portion of flush water transmitted from toilet tank **204**, through fluid pathway **222**, and into rim channel **236** may be less than or equal to 1, 1.2, 1.4, 1.6, 1.8, 2, 2.2, 2.4, 2.6, 2.8, 3, 3.2, 3.4, 3.6, 3.8, 4, 4.2, 4.4, 4.6, 4.8, or 5 liters. Optionally, the portion of flush water transmitted from toilet tank **204**, through fluid pathway **222**, and into rim channel **236** may be greater than or equal to 1, 1.2, 1.4, 1.6, 1.8, 2, 2.2, 2.4, 2.6, 2.8, 3, 3.2, 3.4, 3.6, 3.8, 4, 4.2, 4.4, 4.6, 4.8, or 5 liters.

[0059] In some embodiments, less than 50% of the total volume of flush water delivered from toilet tank **204** to toilet bowl **202** during a flush cycle may be transmitted through rim channel **236**. In some embodiments, greater than 50% of the total volume of flush water delivered from toilet tank **204** to toilet bowl **202** during a flush cycle may be transmitted through rim channel **235**.

[0060] A lower portion of toilet bowl **202**, referred to hereinafter as sump area **214a**, may be in flow communication with a first upstream portion **214u** of trapway **206** through a trapway inlet **207**. The first upstream portion **214u** of trapway **206**, sump area **214a** of toilet bowl **202**, and a first downstream portion **214d** of trapway **206** may together form a sump trap **214**, which may contain flush water between flush cycles.

[0061] In some embodiments, fluid pathway **222** may be in flow communication with a jet channel **238**. Jet channel may be adjacent to an upstream end of trapway **206** and may be fluidically coupled to sump trap **214** via a jet hole **240**. During a flush cycle, a portion of flush water may be transmitted from toilet tank **204**, through fluid pathway **222**, and into jet channel **238**. In some embodiments, the flush water that is transmitted into jet channel **238** may be transferred into sump trap **214** through jet hole **240** in order to generate a siphon in toilet **200**. In some embodiments, the flush water that is transmitted into jet channel **238** may be transferred into sump trap **214** through jet hole **240** at the end of the flush cycle to refill sump trap **214**.

[0062] In some embodiments, the ratio of the volume of flush water transmitted through rim channel **236** during a flush cycle to the volume of flush water transmitted through jet hole **240** during a flush cycle may be approximately 7:1, 6:1, 5:1, 4:1, 3:1, 2:1, 1:1, 1:2, 1:3, 1:4, 1:5, 1:6, or 1:7. Optionally, the ratio of the volume of flush water transmitted through rim channel **236** during a flush cycle to the volume of flush water transmitted through jet hole **240** during a flush cycle may be less than or equal to 7:1, 6:1, 5:1, 4:1, 3:1, 2:1, 1:1, 1:2, 1:3, 1:4, 1:5, 1:6, or 1:7. Alternatively, the ratio of the volume of flush water transmitted through rim channel **236** during a flush cycle to

the volume of flush water transmitted through jet hole **240** during a flush cycle may be greater than or equal to 7:1, 6:1, 5:1, 4:1, 3:1, 2:1, 1:1, 1:2, 1:3, 1:4, 1:5, 1:6, or 1:7.

[0063] Trapway **206** may comprise a first upstream weir **226** configured to contain the water that collects in sump trap **214**. First upstream weir **226** may be a high point of a lower wall of trapway **206** that is downstream of the first downstream portion **214u** of trapway **206** and may be positioned above a highest point of trapway inlet **207**. In some embodiments, first upstream weir **226** may be positioned above the highest point of toilet bowl **202**—trapway **206** interface by an offset distance of about 0.5 inches, about 1 inch, about 1.5 inches, about 2 inches, about 2.5 inches, or about 3 inches. In some embodiments, first upstream weir **226** may be positioned above the highest point of toilet bowl **202**—trapway **206** interface by an offset distance greater than or equal to 2 inches, 2.5 inches, 3 inches, 3.5 inches, or 4 inches. In some embodiments, first upstream weir **226** may be positioned above the highest point of toilet bowl **202**—trapway **206** interface by an offset distance less than or equal to 6 inches, 5.5 inches, 5 inches, 4.5 inches, or 4 inches.

[0064] Trapway **206** may be configured to fluidly connect to a sewer system via a trapway outlet **220**. As a result, sewer gases may enter trapway **206** through trapway outlet **220**. The water that is contained in sump trap **214** between flush cycles may form a water seal in toilet bowl **202**. The water seal may prevent sewer gases which may have entered trapway **206** from leaking out of toilet bowl **202**. Between flush cycles, the water seal in toilet bowl **202** may be vertically lower than first upstream weir **226**. In some embodiments, a vertical distance between the water seal in toilet bowl **202** and first upstream weir **226** may be about 0.1 inches, about 0.2 inches, about 0.3 inches, about 0.4 inches, or about 0.5 inches. In some embodiments, a vertical distance between the water seal in toilet bowl **202** and first upstream weir **226** may be greater than or equal to 0.2 inches, 0.5 inches, 0.8 inches, or 1 inch. In some embodiments, a vertical distance between the water seal in toilet bowl **202** and first upstream weir **226** may be less than or equal to 2 inches, 1.8 inches, 1.5 inches, or 1.2 inches.

[0065] Trapway **206** may comprise a lower trap **216** positioned downstream of first upstream weir **226** between a second upstream portion **216u** of trapway **206** and a second downstream portion **216d** of trapway **206**. Between flush cycles, water may collect in lower trap **216** to form a pair of water seals to provide a second barrier against sewer gases which may enter trapway **206** via trapway outlet **220**. The downstream water seal of the pair of water seals may be vertically higher than an upstream water seal of the pair of water seals. Trapway **206** may comprise a second downstream weir **228** configured to contain water in lower trap **216**. Second downstream weir **228** may be a high point of a lower wall of trapway **206** that is downstream of the second downstream portion **216d** of trapway **206**.

[0066] Container **218** may house a connecting tube **230** configured to provide flow communication between container **218** and trapway **206**. Connecting tube **230** may extend from an interior region of container **218** into trapway **206**. In some embodiments, connecting tube **230** may be coupled to trapway **206** at a position between sump trap **214** and lower trap **216**. A trapway-tank interface **212** may be positioned between sump trap **214** and lower trap **216** and may be configured to couple to connecting tube **230**. In some embodiments, connecting tube **230** may comprise a backflow preventer.

[0067] FIG. 3 shows a perspective cutaway view of a dual-trap toilet, in accordance with some embodiments. Specifically, FIG. 3 illustrates a cross-sectional view of a toilet **300** comprising a toilet bowl **302**, a toilet tank **304**, a trapway **306**, and a flush control **308**. In some embodiments, toilet **300** may include one or more features of toilet **200** shown in FIG. 2 and/or toilet **100** shown in FIGS. 1A and 1B.

[0068] Toilet tank **304** may comprise one or more features of toilet tank **204** of toilet **200** shown in FIG. 2. In some embodiments, toilet tank **304** may be configured to house a container **318**, a flush valve **324**, and a fill valve **334**. Container **318** and flush valve **324** may include one or more features of container **218** and flush valve **224**, respectively. Fill valve **334** may be configured to

fluidly connect to a water supply in order to facilitate the transmission of flush water from the water supply to toilet tank **304**. In some embodiments, fill valve **334** may comprise a backflow preventer.

[0069] Container **318** may have an irregular toroidal (i.e., “donut”-shaped) structure defined by an inner wall **342**, a continuous side wall **346**, and an upper wall **344** that together divide container **318** into an inner chamber **318I** and a vacuum chamber **318V**. In between flush cycles, the flush water in tank **304** may be divided between a region of the toilet tank outside of container **318**, vacuum chamber **318V**, and inner chamber **318I**.

[0070] Vacuum chamber **318V** may have a closed upper end and may be configured to contain pressurized air when toilet **300** is between flush cycles. The pressurized air in vacuum chamber **318V** may have a pressure greater than the ambient pressure of air in the toilet tank. A connecting tube may fluidically couple the upper end of vacuum chamber **318V** to trapway **306**. As a result of the coupling between trapway **306** and the upper end of vacuum chamber **318V**, trapway **306** may also contain pressurized air when the toilet is between flush cycles.

[0071] Inner chamber **318I** may have an upper end that is open to tank **304**, allowing air to flow freely between inner chamber **318I** and the region of the tank that is outside of the container. Inner chamber **318I** may also house flush valve **324**. Upon initiation of a flush cycle, a flush valve seal may be lifted from an inlet to flush valve **324** to open flush valve **324**, and flush water may be discharged from tank **304** and container **318** through flush valve **324** and into toilet bowl **302**. The discharging flush water may reduce the pressure of the air contained in vacuum chamber **318V** of container **318** and air contained in trapway **306** of the toilet. This reduction in pressure may induce a siphon that pulls the contents of toilet bowl **302** into trapway **306** and subsequently into the sewer system.

[0072] Toilet bowl **302** may be fluidly coupled to flush valve **324** via a fluid pathway **322**. A flush valve gasket **310** may form a mechanical seal between flush valve **324** and fluid pathway **322** in order to prevent water from leaking out of toilet tank **304** and into toilet bowl **302** in between flush cycles. In some embodiments, fluid pathway **322** may be in flow communication with a rim channel **336** and/or a jet channel **338**. Rim channel **336** may include one or more features of rim channel **236** shown in FIG. 2. Jet channel **338** may include one or more features of jet channel **238** shown in FIG. 2.

[0073] Trapway **306** may comprise one or more features of trapway **206** of toilet **200** shown in FIG. 2 and/or trapway **106** of toilet **100** shown in FIGS. 1A and 1B. Trapway **306** may comprise a sump trap **314** (defined by a first upstream portion **314u** of trapway **306**, a sump area **314a** in toilet bowl **302**, and a first downstream portion **314d** of trapway **306**), a first upstream weir **326**, a lower trap **316** (positioned between a second upstream portion **316u** of trapway **306** and a second downstream portion **316d** of trapway **306**), and a second downstream weir **328**. Trapway **306** may be fluidly coupled to toilet tank **304** via a trapway-tank interface **312**. In some embodiments, trapway **306** may comprise an outlet **320** configured to fluidly connect to a drainage pipe into a sewer system.

[0074] FIG. 4 shows a partial cross-sectional view of a dual-trap toilet, in accordance with some embodiments. Specifically, FIG. 4 shows a partial cross-sectional view of a toilet **400**. Toilet **400** may comprise a toilet bowl **402** that is fluidly coupled to a toilet tank via a flush valve outlet **404**. Toilet bowl **402** may be fluidly coupled to a trapway **410** through a trapway inlet **411**. A sump trap **412** may be defined by a first upstream portion **412u**, a sump area **412a**, and a first downstream portion **412d** of trapway **410**. A first upstream weir **416** of trapway **410** may be positioned downstream of sump trap **412**. A lower trap **414** may be positioned between a second upstream portion **414u** and a second downstream portion **414d** of trapway **410**, and a second downstream weir **418** may be positioned downstream of lower trap **414**. Trapway **410** may be fluidly coupled to a toilet tank via a trapway-tank interface **408**. In some embodiments, one or more features of toilet **400** may include one or more features of toilet **300** shown in FIG. 3, toilet **200** shown in FIG. 2,

and/or toilet **100** shown in FIGS. **1A-1B**.

Connecting Tube Interface

[0075] The trapway of a dual-trap toilet may be fluidly coupled to a connecting tube that is positioned within the toilet tank. The connecting tube may provide flow communication between a toilet tank container and a portion of the trapway that contains a volume of pressurized air. Similarly, the toilet bowl of the dual-trap toilet may be fluidly coupled to a flush valve in the toilet tank. Upon initiation of a flush cycle, flush water may be discharged into the toilet bowl via the flush valve in order to cause the contents of the toilet bowl to be pulled into the trapway. In some embodiments, the dual-trap toilet may have one or more gaskets that are configured to provide an effective seal between components in the toilet tank and the trapway and/or the toilet bowl.

[0076] FIGS. **5A-5C** show various views of components of a dual-trap toilet for connecting a connecting tube between a toilet tank and a trapway, in accordance with some embodiments. Specifically, FIGS. **5A-5C** show various views of a toilet deck **500** comprising a tank-trapway interface **502**, a flush water outlet **504**, a connecting tube coupling **506**, and a flush water flow path **508**. Tank-trapway interface **502** may be fluidly coupled to a connecting tube **514** in order to provide flow communication between a first volume of pressurized air that is contained in an upper portion of a container **518** in the toilet tank and a second volume of pressurized air that is contained in the trapway. Flush water outlet **504** may be fluidly coupled to a flush valve assembly that is positioned in the toilet tank. Flush water flow path **508** may be configured to transmit flush water that is discharged by a flush valve to a toilet bowl of the dual-trap toilet.

[0077] A first gasket **510** may be positioned between the toilet tank and toilet deck **500**.

Specifically, first gasket **510** may be positioned between a lower end of the flush valve assembly that is housed in the toilet tank and flush water outlet **504**. In some embodiments, first gasket **510** may provide a seal around a flush valve assembly by forming a seal around a portion of the flush valve assembly through which flush water from the flush valve may pass from the toilet tank into the toilet bowl. The seal may prevent flush water from leaking into the toilet bowl between flush cycles. First gasket **510** comprise one or more flexible and/or compressible materials, such as rubber or any natural or synthetic elastomer. In some embodiments, first gasket **510** may have a foam specification 1A3.

[0078] A second gasket **512** may be positioned between the toilet tank and toilet deck **500**.

Specifically, second gasket **512** may be positioned between a lower end of connecting tube **514** and connecting tube coupling **506**. In some embodiments, second gasket **512** may provide a seal around connecting tube **514** by forming a seal around a portion of the conduit through which pressurized air may pass between container **518** and the trapway. The seal may prevent pressurized air and/or vacuum from leaking. The seal that is formed between connecting tube **514**, the toilet tank, and toilet deck **500** may allow for a vacuum to be held in an upper end of container **518**, in connecting tube **514**, and/or in the trapway during flush cycles. In some embodiments, the seal may allow for pressurized air to be held in the upper end of container **518**, in connecting tube **514**, and/or in the trapway between flush cycles. Second gasket **512** comprise one or more flexible and/or compressible materials, such as rubber or any natural or synthetic elastomer. In some embodiments, second gasket **512** may have a Durometer softness of 20-60 Shore A.

[0079] First gasket **510** and second gasket **512** may be provided separately from one another.

Providing first gasket **510** and second gasket **512** separately from one another, rather than providing a single gasket with multiple through-holes intended to provide both seals, may allow for the preciseness of the fit and the effectiveness of the seal of each of the gaskets to be improved.

[0080] Close-up view of first gasket **510** and second gasket **512** are provided in FIGS. **5D-5E**. First gasket **510** and/or second gasket **512** may have a generally annular shape that encircles a central flow path through which water and/or air flows. In some embodiments, rather than being provided in the form of a conventional o-ring, first gasket **510** and/or second gasket **512** may be shaped to provide an improved seal that is more tightly and precisely fitting, is easier to install, and is more

robust against leaks. In some embodiments, first gasket **510** may comprise a first recess and/or protrusion **522** configured to receive the flush valve assembly and/or to protrude into toilet deck **500**. In some embodiments, **512** may comprise a second recess and/or protrusion **526** configured to receive connecting tube **514** and/or to protrude into toilet deck **500**. In some embodiments, a flipped configuration may be used, for example in which a protrusion extends upwards into the toilet tank and a recess receives a component from below.

[0081] In some embodiments, first gasket **510** may include a first collar portion **520** formed around first recess **522**, wherein the outer circumference (and/or diameter) of first collar portion **520** may be larger than a circumference (and/or diameter) of first recess and/or protrusion **522**. First collar portion **520** may have a circumference (and/or diameter) that is larger than the outer circumference (and/or diameter) of first recess and/or protrusion **522**. First collar portion **520** may have a greater height (e.g., in the flow direction through the gasket), a lesser height, or the same height as the height of first recess and/or protrusion **522** formed in first gasket **510**. First collar portion **520** may have a greater height (e.g., in the flow direction through the gasket), a lesser height, or the same height as the height of recess and/or protrusion **522**. In some embodiments, an inner circumference (and/or diameter) of first recess and/or protrusion **522** may be greater, lesser, or equal to the outer circumference (and/or diameter) of first recess and/or protrusion **522**. An upper portion of first gasket **510** may be configured to receive a hex nut.

[0082] In some embodiments, second gasket **512** may include a second collar portion **524** formed around second recess/protrusion **526**, wherein the outer circumference (and/or diameter) of second collar portion **524** may be larger than a circumference (and/or diameter) of second recess and/or protrusion **526**. Second collar portion **524** may have a circumference (and/or diameter) that is larger than the outer circumference (and/or diameter) of second recess and/or protrusion **526**. Second collar portion **524** may have a greater height (e.g., in the flow direction through the gasket), a lesser height, or the same height as the height of second recess and/or protrusion **526** formed in second gasket **512**. Second collar portion **524** may have a greater height (e.g., in the flow direction through the gasket), a lesser height, or the same height as the height of second recess and/or protrusion **526**. In some embodiments, an inner circumference (and/or diameter) of second recess and/or protrusion **526** may be greater, lesser, or equal to the outer circumference (and/or diameter) of second recess and/or protrusion **526**. An upper portion of second gasket **512** may be configured to receive a hex nut.

[0083] FIGS. **6A-6D** show various cross-sectional views of an interface **600** between a toilet tank **602** and a toilet deck **604**, showing a valve gasket and a vacuum gasket, in accordance with some embodiments. An exploded view of interface **600** is shown in FIG. **6E**. Toilet tank **602** may house a flush valve assembly **606** and a connecting tube **610**. In some embodiments, connecting tube **612** may be housed within a container **628** that is positioned in toilet tank **602**. In some embodiments, toilet deck **604** may comprise a flush water outlet **610** configured to receive an opening in flush valve assembly **606**. Flush water outlet may be fluidically coupled to a fluid pathway/manifold **636** (which may be similar to or include one or more features of fluid pathway **322** shown in FIG. **3**). In some embodiments, toilet deck **604** may comprise a connecting tube coupling **616** configured to receive an opening in connecting tube **612**.

[0084] A first gasket **608** may be positioned between a lower end of flush valve assembly **606** and flush water outlet **610**. A second gasket **614** may be positioned between a lower end of connecting tube **612** and connecting tube coupling **616**. First gasket **608** may include one or more features of first gasket **510** shown in FIG. **5**. Second gasket **614** may include one or more features of second gasket **512** shown in FIG. **5**. In particular, first gasket **608** and second gasket **614** may comprise collared portions and recessed and/or protrusion portions. The recesses and/or protrusions may allow for first gasket **608** and/or second gasket **614** to be compressed by forces acting in multiple different directions, for example including a vertical direction and a horizontal direction as explained below. Compression by forces acting in multiple different directions may allow for a

tighter and more secure fit to be achieved, and for a more effective and robust seal to be formed. A first nut **632** and a second nut **630** may be disposed in an upper portion of first gasket **608** and second gasket **614**, respectively.

[0085] Second gasket **614** may comprise vacuum gasket seals at multiple points to allow for repeatable, stable vacuum formation without loss of vacuum. A first vacuum gasket seal may be a face seal **618**. Face seal **618** may vertically compress a collared portion of second gasket **614** between toilet tank **602** and toilet deck **604**. A second vacuum gasket seal **620** may horizontally compress a recessed portion of second gasket **614** between a lower portion of connecting tube **610** and connecting tube coupling **616**. A third vacuum gasket seal may be a piston seal **622**. Piston seal **622** may horizontally or radially compress a recessed/protruding portion of second gasket **614** between a lower portion of connecting tube **610** and a terminal portion of connecting tube coupling **616**. In some embodiments, first gasket **608** may comprise one or more gasket seals analogous to seals **618-622**, for example a face seal and/or a piston seal.

[0086] A third gasket **624** may be positioned around flush valve assembly **606**. Third gasket **624** may be coupled to container **628** and may, in some embodiments, be configured to sit on or near a bottom surface of an inner portion of tank **602**. In some embodiments, third gasket **624** may be configured to facilitate installation. For example, third gasket **624** may be symmetrical in shape (e.g., symmetric about a central plane that is level with the bottom surface of tank **602** when third gasket **624** is installed) to ensure ease of installation. Third gasket **624** may be annular in shape and may comprise a central through-hole through which a lower portion of flush valve assembly is configured to extend. An outer edge of third gasket **624** may comprise an upper lip configured to extend over a top, internal surface of the bottom of container **628** and a bottom lip configured to extend underneath a bottom, external surface of the bottom of container **628**.

[0087] A fourth gasket **626** may be positioned between connecting tube **612** and a bottom surface of an inner portion of tank **602**. Fourth gasket **626** may be configured to prevent flush water from leaking out of tank **602**. Fourth gasket **626** may be annular in shape and may comprise a central through-hole through which connecting tube **612** is configured to extend. An outer edge of gasket **626** may comprise a lip portion that extends laterally away from connecting tube **612** along a bottom, internal surface of tank **602**.

[0088] In some embodiments, as shown in FIGS. **6B-6E**, interface **600** may include a manifold plate **634**. Manifold plate **634** may comprise a plurality of legs that rest on a bottom surface of tank **602** between flush valve **606** and fluid pathway/manifold **634**. The position of manifold plate **634** in tank **602** may be fixed by a plurality of bolts **638** and a plurality of rubber grommets **640**.

Manifold plate **634** may compress gaskets **608**, **614**, **624**, **626**, and/or a donut gasket **642** to prevent vacuum and water leaks. A top view and a bottom view of an exemplary manifold plate are shown in FIG. **7A** and FIG. **7B**, respectively.

Exemplary Embodiments

[0089] Below is an enumerated listing of certain embodiments. In some embodiments, any one or more features of any one or more of the embodiments below may be combined with any one or more of the other embodiments, even if the dependencies of the embodiments do not explicitly indicate that the embodiments may be combined in such manner. In some embodiments, any one or more of the features of any one or more of the embodiments below may be combined with any one or more features or aspects otherwise disclosed in this application. [0090] 1. A toilet assembly, comprising: [0091] a toilet tank to hold flush water; [0092] a flush valve assembly positioned in the toilet tank; [0093] a toilet deck; [0094] a toilet bowl; [0095] a trapway in flow communication with the toilet bowl; [0096] a container positioned in the toilet tank; [0097] a connecting tube extending from an interior of the container to the trapway, [0098] a first gasket positioned between the toilet tank and the toilet deck and forming a seal around the flush valve assembly; and [0099] a second gasket positioned between the toilet tank and the toilet deck and forming a seal around the connecting tube; [0100] wherein: [0101] the container is in flow communication with the toilet

tank, the container having an open lower end and a closed upper end, [0102] the trapway comprises a sump trap, a first upstream weir, a lower trap, and a second downstream weir, and [0103] the connecting tube is coupled to the trapway at a position between the sump trap and the lower trap and provides flow communication between the container and the trapway. [0104] 2. The toilet assembly of embodiment 1, wherein the first gasket comprises a recess receiving the flush valve assembly. [0105] 3. The toilet assembly of any one of embodiments 1-2, wherein the first gasket comprises a protrusion extending into the toilet deck. [0106] 4. The toilet assembly of embodiment 3, wherein the first gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion. [0107] 5. The toilet assembly of any one of embodiments 1-4, wherein the second gasket comprises a recess receiving the connecting tube. [0108] 6. The toilet assembly of any one of embodiments 1-5, wherein the second gasket comprises a protrusion extending into the toilet deck. [0109] 7. The toilet assembly of embodiment 6, wherein the second gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion. [0110] 8. The toilet assembly of any one of embodiments 1-7, wherein, between flush cycles, the container comprises pressurized air positioned above flush water. [0111] 9. The toilet assembly of any one of embodiments 1-8, wherein, upon discharging water into a flush valve core to initiate a flush cycle, reduced pressure is created in pressurized air in the trapway. [0112] 10. The toilet assembly of any one of embodiments 1-9, wherein, upon a toilet tank water level falling to a lower edge of a head of the flush valve to break a siphon, and re-filling the toilet tank with water to end a flush cycle, increased pressure is created in pressurized air in the trapway. [0113] 11. The toilet assembly of any one of embodiments 1-10, wherein, between flush cycles, pressurized air in the trapway is pressurized at a pressure greater than or equal to 0.5 cm of water above atmospheric pressure and less than or equal to 5 cm of water above atmospheric pressure. [0114] 12. The toilet assembly of any one of embodiments 1-11, wherein the first upstream weir is positioned above a highest point of a trapway inlet by an offset distance of greater than or equal to 2 inches. [0115] 13. The toilet assembly of any one of embodiments 1-12, wherein the first upstream weir is positioned at point that, between flush cycles, is vertically higher than a water seal of the toilet bowl by a distance of greater than or equal to 0.2 inches. [0116] 14. The toilet assembly of any one of embodiments 1-13, wherein the connecting tube comprises a backflow preventer. [0117] 15. The toilet assembly of any one of embodiments 1-14, wherein the flush valve assembly comprises a valve body extending from a valve inlet to a valve outlet and a valve cover having a seal to enclose the valve inlet. [0118] 16. The toilet assembly of any one of embodiments 1-15, wherein the container comprises a continuous side wall, an upper end wall, and an inner wall that together bound a vacuum chamber of the container. [0119] 17. The toilet assembly of embodiment 16, wherein the inner wall bounds an inner chamber of the container. [0120] 18. A toilet assembly, comprising [0121] a toilet tank to hold flush water; [0122] a flush valve assembly positioned in the toilet tank; [0123] a toilet deck; [0124] a toilet bowl; [0125] a trapway in flow communication with the toilet bowl; [0126] a container positioned in the toilet tank; [0127] a connecting tube extending from an interior of the container to the trapway, and [0128] a first gasket positioned between the toilet tank and the toilet deck and forming a seal around the flush valve assembly, wherein the first gasket comprises a recess receiving the flush valve assembly and a protrusion extending into the toilet deck; [0129] wherein: [0130] the container is in flow communication with the toilet tank, the container having an open lower end and a closed upper end, [0131] the trapway comprises a sump trap, a first upstream weir, a lower trap, and a second downstream weir, and [0132] the connecting tube is coupled to the trapway at a position between the sump trap and the lower trap and provides flow communication between the container and the trapway. [0133] 19. The toilet assembly of embodiment 18, wherein the first gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion. [0134] 20. The toilet assembly of any one of embodiment 18-19, wherein an outer diameter of the protrusion of the first gasket is wider than an inner diameter of the recess of the first gasket. [0135] 21. The toilet assembly of any one of embodiments 18-20,

comprising a second gasket positioned between the toilet tank and the toilet deck and forming a seal around the connecting tube, wherein the second gasket comprises a recess receiving the connecting tube and a protrusion extending into the toilet deck. [0136] 22. The toilet assembly of embodiment 21, wherein the second gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion of the second gasket. [0137] 23. The toilet assembly of any one of embodiments 21-22, wherein an outer diameter of the protrusion of the second gasket is wider than an inner diameter of the recess of the second gasket. [0138] 24. The toilet assembly of any one of embodiments 18-23, wherein, between flush cycles, the container comprises pressurized air positioned above flush water. [0139] 25. The toilet assembly of any one of embodiments 18-24, wherein, upon discharging water into a flush valve core to initiate a flush cycle, reduced pressure is created in pressurized air in the trapway. [0140] 26. The toilet assembly of any one of embodiments 18-25, wherein, upon a toilet tank water level falling to a lower edge of a head of the flush valve to break a siphon, and re-filling the toilet tank with water to end a flush cycle, increased pressure is created in pressurized air in the trapway. [0141] 27. The toilet assembly of any one of embodiments 18-26, wherein, between flush cycles, pressurized air in the trapway is pressurized at a pressure greater than or equal to 0.5 cm of water above atmospheric pressure and less than or equal to 5 cm of water above atmospheric pressure. [0142] 28. The toilet assembly of any one of embodiments 18-27, wherein the first upstream weir is positioned above a highest point of a trapway inlet by an offset distance of greater than or equal to 2 inches. [0143] 29. The toilet assembly of any one of embodiments 18-28, wherein the first upstream weir is positioned at point that, between flush cycles, is vertically higher than a water seal of the toilet bowl by a distance of greater than or equal to 0.2 inches. [0144] 30. The toilet assembly of any one of embodiments 18-29, wherein the connecting tube comprises a backflow preventer. [0145] 31. The toilet assembly of any one of embodiments 18-30, wherein the flush valve assembly comprises a valve body extending from a valve inlet to a valve outlet and a valve cover having a seal to enclose the valve inlet. [0146] 32. The toilet assembly of any one of embodiments 18-31, wherein the container comprises a continuous side wall, an upper end wall, and an inner wall that together bound a vacuum chamber of the container. [0147] 33. The toilet assembly of embodiment 32, wherein the inner wall bounds an inner chamber of the container. [0148] 34. The toilet assembly of any one of embodiments 1-17 or 21, wherein the second gasket comprises a face seal. [0149] 35. The toilet assembly of any one of embodiments 1-17 or 21, wherein the second gasket comprises a piston seal.

Conclusion

[0150] In some embodiments, dual-trap toilets described and/or shown herein may share any one or more characteristics in common with dual-trap toilets described, shown, and/or referenced in PCT publication no. WO2022046876A1, in PCT publication no. WO2022051331A1, and/or in PCT application no. PCT/US2021/060307 (filed Nov. 22, 2021), the entirety of each of which is hereby incorporated by reference.

[0151] Any one or more characteristics of any of the embodiments (including claims) described, shown, and/or referenced herein may be combined, in whole or in part, with any one or more characteristics of any one or more other embodiments (including claims) described, shown, and/or referenced herein.

[0152] The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated.

[0153] Although the disclosure and examples have been fully described with reference to the accompanying figures, it is to be noted that various changes and modifications will become

apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims. Finally, the entire disclosure of the patents and publications referred to in this application are hereby incorporated herein by reference.

Claims

1. A toilet assembly, comprising: a toilet tank to hold flush water; a flush valve assembly positioned in the toilet tank; a toilet deck; a toilet bowl; a trapway in flow communication with the toilet bowl; a container positioned in the toilet tank, and a connecting tube extending from an interior of the container to the trapway, a first gasket positioned between the toilet tank and the toilet deck and forming a seal around the flush valve assembly; and a second gasket positioned between the toilet tank and the toilet deck and forming a seal around the connecting tube; wherein: the container is in flow communication with the toilet tank, the container having an open lower end and a closed upper end, the trapway comprises a sump trap, a first upstream weir, a lower trap, and a second downstream weir, the connecting tube is coupled to the trapway at a position between the sump trap and the lower trap, and provides flow communication between the container and the trapway.
2. The toilet assembly of claim 1, wherein the first gasket comprises a recess receiving the flush valve assembly.
3. The toilet assembly of claim 1, wherein the first gasket comprises a protrusion extending into the toilet deck.
4. The toilet assembly of claim 3, wherein the first gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion.
5. The toilet assembly of claim 1, wherein the second gasket comprises a recess receiving the connecting tube.
6. The toilet assembly of claim 1, wherein the second gasket comprises a protrusion extending into the toilet deck.
7. The toilet assembly of claim 6, wherein the second gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion.
8. The toilet assembly of claim 1, wherein, between flush cycles, the container comprises pressurized air positioned above flush water.
9. The toilet assembly of claim 1, wherein, upon discharging water into a flush valve core to initiate a flush cycle, reduced pressure is created in pressurized air in the trapway.
10. The toilet assembly of claim 1, wherein, upon re-filling the toilet tank with water to end a flush cycle, increased pressure is created in pressurized air in the trapway.
11. The toilet assembly of claim 1, wherein the flush valve assembly comprises a valve body extending from a valve inlet to a valve outlet and a valve cover having a seal to enclose the valve inlet.
12. The toilet assembly of claim 1, wherein the container comprises a continuous side wall, an upper end wall, and an inner wall that together bound a vacuum chamber of the container.
13. The toilet assembly of claim 12, wherein the inner wall bounds an inner chamber of the container.
14. The toilet assembly of claim 1, wherein the second gasket comprises a face seal.
15. The toilet assembly of claim 1, wherein the second gasket comprises a piston seal.
16. A toilet assembly, comprising a toilet tank to hold flush water; a flush valve assembly positioned in the toilet tank; a toilet deck; a toilet bowl; a trapway in flow communication with the toilet bowl; a container positioned in the toilet tank, and a connecting tube extending from an interior of the container to the trapway, a first gasket positioned between the toilet tank and the toilet deck and forming a seal around the flush valve assembly, wherein the first gasket comprises a recess receiving the flush valve assembly and a protrusion extending into the toilet deck; and

wherein: the container is in flow communication with the toilet tank, the container having an open lower end and a closed upper end, the trapway comprises a sump trap, a first upstream weir, a lower trap, and a second downstream weir, the connecting tube is coupled to the trapway at a position between the sump trap and the lower trap, and provides flow communication between the container and the trapway.

17. The toilet assembly of claim 16, wherein the first gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion.

18. The toilet assembly of claim 16, wherein an outer diameter of the protrusion of the first gasket is wider than an inner diameter of the recess of the first gasket.

19. The toilet assembly of claim 16, comprising a second gasket positioned between the toilet tank and the toilet deck and forming a seal around the connecting tube, wherein the second gasket comprises a recess receiving the connecting tube and a protrusion extending into the toilet deck.

20. The toilet assembly of claim 19, wherein the second gasket comprises a collar portion having a wider diameter than an outer diameter of the protrusion of the second gasket.

21. The toilet assembly of claim 19, wherein an outer diameter of the protrusion of the second gasket is wider than an inner diameter of the recess of the second gasket.

22. The toilet assembly of claim 19, wherein the second gasket comprises a face seal.

23. The toilet assembly of claim 19, wherein the second gasket comprises a piston seal.
